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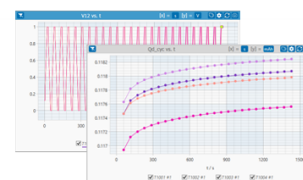
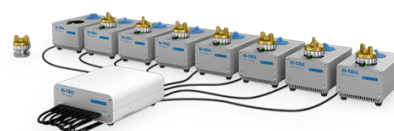
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The Impact of Upper Cut-Off Voltage on the Cycling Performance of Li-Ion Cells with Positive Electrodes Having Various Nickel Contents

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The charge-discharge cycling performance of pouch cells with single crystal $\text{LiNi}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2}\text{O}_2$ (SC532), $\text{LiNi}_{0.8}\text{Mn}_{0.1}\text{Co}_{0.1}\text{O}_2$ (SC811) and a prototype polycrystalline Co-free core-shell material with an average 94% Ni content (Ni94) were compared in this work. Two upper cut-off voltages (UCVs) per cell type were chosen to either include or exclude the remnant of the “H2-H3 phase transition” region, if present, of each positive electrode material. The core-shell Ni94 shows comparable performance to the SC532 and better performance than the SC811 only at 20 °C and 4.04 V UCV. In other testing conditions, the SC532 has the best performance followed by the SC811. The cross-section SEM images of the fresh Ni94 electrode show microcracks from electrode calendaring which is detrimental to its cycling performance as the exposed Ni-rich core has a high reactivity with the electrolyte which induces large impedance increase. The Ni94 material shows quite poor capacity retention and large impedance growth when charged to 4.18 V, through the large volume change associated with the “H2-H3 remnant,” but acceptable capacity retention when only charged to 4.04 V that avoids this large volume change.

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Further improving the energy density of lithium-ion cells is one of the main research directions in both academia and industry. The most straightforward method is to simply increase the upper cut-off voltages (UCVs) of a lithium-ion cell with sufficient graphite provided. By carefully choosing electrolyte blends, side reactions between the positive electrode materials and the electrolyte can be lowered so that a lithium-ion cell can obtain acceptable lifetime when charged to a high voltage.^{1,2}

The use of Ni-rich (>90%) low/zero Co layered metal oxide positive electrode materials provides another option to further increase the energy density as well as lower the cost of lithium-ion cells. However, though Ni-rich materials bring higher specific capacity than their low Ni content counterparts, the capacity retention and lifetime of Li-ion cells with Ni-rich materials generally appears to be inferior.^{3,4} From the material structural perspective, many reports have related this observation to the so-called “H2-H3 phase transition” originating from the study of the LiNiO_2 , where this two-phase region appears at around 4.2 V vs Li/Li^+ in the dQ/dV vs V curve.^{5–11} However, H. Li et al.¹² have shown through in situ XRD that doping LiNiO_2 with only 5% of Mg ($\text{LiNi}_{0.95}\text{Mg}_{0.05}\text{O}_2$), Al ($\text{LiNi}_{0.95}\text{Al}_{0.05}\text{O}_2$) or Mn ($\text{LiNi}_{0.95}\text{Mn}_{0.05}\text{O}_2$) results in a single-phase material, even though the dQ/dV vs V curves of these materials show prominent peaks at around 4.2 V vs Li/Li^+ . This shows that the dQ/dV peaks do not necessarily correspond to a two-phase region as in the case of pure LiNiO_2 . The dQ/dV peaks at around 4.2 V vs Li/Li^+ for $\text{LiNi}_{0.95}\text{M}_{0.05}\text{O}_2$ (M = Mg, Al and Mn) are a “remnant” of the H2-H3 phase transition in LiNiO_2 and correspond to a region where the unit cell volume of the electrode material contracts rapidly as lithium is removed. When cycling through this “remnant” peak, lattice contraction of Ni-rich materials becomes severe due to high degree of delithiation and microcracks can be developed for

polycrystalline materials.^{13,14} These microcracks are believed to cause active materials to lose electric-connection and allow electrolyte infiltration into particle interiors inducing more side reactions that increase cell impedance.^{15,16} Because of this, it has been shown that the single crystal materials such as $\text{LiNi}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2}\text{O}_2$ (SC532) and $\text{LiNi}_{0.8}\text{Mn}_{0.1}\text{Co}_{0.1}\text{O}_2$ (SC811) perform significantly better than their polycrystalline counterparts in commercial-grade pouch cells.^{2,17}

To explore the impact of the “H2-H3 phase transition remnant” (“H2-H3 remnant” for short) on the cycling performance, pouch cells using commercially available SC532 and SC811, along with a prototype polycrystalline Ni-rich Co-free core-shell material with an average of 94% Ni content (Ni94) obtained from reputable supplier, were the focus of this study. The Ni94 material produced at laboratory-scale was described in detail previously.^{18,19} Unlike in previous studies where the materials were evaluated in laboratory-scale test vehicles,^{18–20} the SC532, SC811 and the prototype core-shell Ni94 material were evaluated in commercial-grade pouch cells with artificial graphite negatives. The UCVs of these full cells were carefully selected to either include or exclude the “H2-H3 phase transition” remnant, when present. Additionally, several previously reported electrolyte blends used in the SC532 and SC811 pouch cells were also examined in the prototype polycrystalline Co-free core-shell Ni94 pouch cells.^{2,21–23}

Experimental

Positive electrode material.—Coated single crystal SC532 was provided by Vendor 1 and is the same material feature in Ref. 24. Uncoated single crystal SC811 was provided by Vendor 2 and is the same material featured in Ref. 15. The polycrystalline Co-free core-shell material has an average composition of $\text{LiNi}_{0.94}\text{Mn}_{0.06}\text{O}_2$. The core-shell hydroxide precursor has a 16 μm $\text{Ni}(\text{OH})_2$ core and a 1 μm thick $\text{Ni}_{0.8}\text{Mn}_{0.2}(\text{OH})_2$ shell and was obtained from Hunan Zoomwe Zhengyuan Advanced Material Trade Company, Ltd., Changsha 410000, China. The final core-shell oxide was prepared by reacting $\text{LiOH}\cdot\text{H}_2\text{O}$ with the hydroxide precursor at elevated temperature by vendor 3. There was no coating of any type

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applied to this material. The core-shell powder was processed by the standard industrial processes including material washing to remove surface lithium residuals. This was the core-shell material with the same composition prepared at laboratory-scale reported previously.^{18,19}

Coin cell half-cell testing and in situ X-ray diffraction (XRD).—Electrodes used for SC532 and SC811 coin cell measurements were directly punched from the single-side-coated parts of the pouch cell electrodes. The electrode loadings in the SC532 and SC811 coin cells were around 21 mg cm⁻². The electrode used for the prototype core-shell Ni94 coin cell was hand-made according to the methods in Ref. 25 from the Ni94 powder provided by Vendor 3. The electrode loading was around 10 mg cm⁻². All electrodes had a diameter of 1.28 cm and were further dried under vacuum at 120 °C for 14 h before coin cell building. 1.2 M LiPF₆ in 1:4 (v/v) fluoroethylene carbonate: dimethyl carbonate electrolyte was used for the half-cell tests. Standard 2325 coin cells were assembled in an argon-filled glovebox. Each coin cell had one positive electrode and one piece of Li foil negative electrode with 2 pieces of Celgard 2300 separator. Galvanostatic charge/discharge cycling was conducted with an E-one Moli Energy Canada battery testing system. Coin cells were cycled between 3 and 4.4 V at 30 °C with 11 mA g⁻¹ current.

Two extra coin cell half-cells of the SC811 and the core-shell Ni94 materials were made and were first cycled with two initial C/20 cycles followed sequentially by 50 cycles of C/5 and 2 cycles of C/20 from 3 to 4.3 V at 30 °C. The electrodes of these coin cells were also punched from the single-side-coated part of the pouch cells.

The in situ XRD studies of SC532, SC811 and LiNi_{0.95}Mn_{0.05}O₂ (not to be confused with the prototype core-shell Ni94), and the following unit cell volume determination vs voltage or lithium content were carried out in previous work.^{12,26}

Pouch cell formation and electrolyte.—The 402035 size machine-made pouch cells were built with Kaijin AML400 (China) artificial graphite (AG) negatives by LiFun Technology (Xinma Industry Zone, Golden Dragon Road, Tianyuan District, Zhuzhou City, Hunan Province, PRC, 412000). The pouch cells were built for the first time for the prototype Co-free core-shell Ni94 material. All pouch cells were vacuum sealed in a dry room without electrolyte and then shipped to Dalhousie University in Canada. Pouch cell specifications are shown in Table I.

The formation for the SC532 pouch cells was described in many previously published papers.^{21,22,27} The SC532 pouch cells were formed at 40 °C under C/20 CCCV mode (constant current charging to top of charge followed by constant voltage hold) to either 4.1 V or 4.3 V. CV time was set to be 1 h. The cells were then discharged to 3.8 V for gas and impedance measurement.

The formation for the SC811 pouch cells was described in the paper published by W. Song et al.² The SC811 pouch cells were formed at 40 °C under C/20 CCCV mode to either 4.1 V or 4.2 V. CV time was set to be 1 h. The cells were then discharged to 3.8 V for gas and impedance measurement.

The prototype core-shell Ni94 pouch cells were formed by one full C/20 CC cycle to either 4.1 V or 4.3 V at 40 °C with a 3.0 V lower voltage cut-off. The cells were then charged to 3.8 V for gas and impedance measurements.

The electrolyte blends used in each pouch cell are based on our previous experience on SC532 and SC811 materials.^{2,21,22,27} Electrolyte blends which yield long lifetimes were selected. The solvent blend used in this work was ethylene carbonate: ethyl methyl carbonate: dimethyl carbonate 25:5:75 by volume. All electrolytes used were either 1.2 M or 1.5 M LiPF₆ salt from either BASF or Shenzhen Capchem. A control electrolyte is a blend of the solvent and LiPF₆ salt only.

Electrolyte additives were added to the control electrolyte with respect to the total electrolyte weight:

1. 2% VC (vinylene carbonate) and 1% DTD (ethylene sulfite), abbreviated as 2VC1DTD, promoted by J. Li et al.,²¹
2. 2% FEC (fluoroethylene carbonate) and 1% LFO (LiPO₂F₂ - lithium difluorophosphate), abbreviated as 2FEC1LFO, promoted by L. Ma et al.,²³
3. 1% LiPO₂F₂, abbreviated as 1LFO, promoted by L. Ma et al.²⁷
4. 1% LFO and 1% ODT (1,2,6-Oxadithiane 2,2,6,6-tetra-oxide), abbreviated as 1LFO1ODT, promoted by X. Ma et al.²²

Gas and impedance measurements.—The gas measurement was conducted using Archimedes' principle.²⁸ Briefly, the weight of each pouch cell submerged in DI water was measured before/after formation. The weight difference is proportional to the gas volume generated.

The impedance was measured at 3.8 V via electrochemical impedance spectroscopy on a Biologic VMP-3 instrument. Pouch cells were allowed to thermally equilibrate for 1 h at 10 °C before measurement. Alternating current impedance spectra were collected with ten points per decade from 100 kHz to 10 mHz with a signal amplitude of 10 mV. The charge-transfer resistance R_{ct} is defined herein as the span of the convoluted "semicircles" from high to low frequency for simplicity of discussion.

Pouch cell cycling.—Two testing protocols were used in this study on Neware (Shenzhen, China) chargers.

In the protocol for long term cycling measured at 20 and 40 °C, two upper cut-off voltages (UCVs) were selected for each positive material. The SC532 cells were cycled from 3.0 to either 4.1 V or 4.3 V. The SC811 cells were cycled from 3.0 to either 4.06 V or 4.2 V. The core-shell Ni94 cells were cycled from 3.0 to either 4.04 V or 4.18 V. The pouch cell voltage polarization is defined as the average voltage difference between the charge and discharge curves of the same cycle. The capacity and voltage polarization of the 5th cycle was taken as 100%. The SC532 and SC811 cells were measured with C/3 CCCV mode at 40 °C and 1C CCCV mode at 20

Table I. Electrode specifications of SC532/AG, SC811/AG, washed core-shell/AG, and core-shell Ni94/AG pouch cells.

		SC532	SC811	Core-shell Ni94
Positive Electrode	Loading [mg cm ⁻²]	21	21	19
	Active material fraction [%]	94	94	94
	Density [g cm ⁻³]	3.5	3.5	3.2
	Reversible capacity [mAh g ⁻¹]	180	210	220
	Total coating mass [g]	1.5	1.2	1.06
Negative electrode	Loading [mg cm ⁻²]	12	15	14
	Active material fraction [%]	95.4	95.4	95.4
	Density [g cm ⁻³]	1.55	1.55	1.5
	Reversible capacity [mAh g ⁻¹]	350	350	350
	Total coating mass [g]	0.9	1	0.9

°C. All the core-shell Ni94 cells were measured with C/3 CCCV mode at both 20 °C and 40 °C.

The 2nd protocol was adopted only for the prototype Co-free core-shell Ni94 pouch cells with 2FEC1LFO additive measured at 20 °C and 40 °C, C/3 CCCV mode. The pouch cell was cycled with alternating UCVs of 4.18 V and 4.04 V. Each UCVs were alternated every 10 cycles.

Ion-milling and cross-section scanning electron microscopy (SEM).—Cross-section SEM images of the Co-free core-shell Ni94 positive electrode were taken from both the fresh and cycled pouch cells. Cells after long term cycling were slowly charged to top of charge and opened carefully in an argon-filled glove box to extract the cycled positive electrode. Cycled electrodes were rinsed with dimethyl carbonate twice before ion-milling with a JEOL IB-19530CP cross-section polisher. All electrodes were milled with an Ar-ion beam for a 90 min and 6 kV coarse step followed by a 5 min and 4 kV fine step. The SEM images were taken using a Hitachi S-4700 field emission electrode microscope with a secondary electron detector. The images were obtained using an accelerating voltage of 5 kV and current of 15 μ A.

Results and Discussion

Figure 1a shows the voltage (vs Li/Li⁺) vs specific capacity of the SC532, SC811 and the prototype core-shell Ni94 positive electrode materials measured at 30 °C. Increasing the Ni content increases the first charge capacity as expected. The irreversible capacity depends on a number of factors such as the Li-ion diffusion constant at low potentials, the discharge rate and the primary particle size.^{29–31} The diffusion constant is affected by the material composition and also by the fraction of Ni atoms in the Li layer.

There is no simple relationship between material composition and irreversible capacity.

Figure 1b shows the dQ/dV curves (left y-axis) and the unit cell volumes (right y-axis) of the SC532, SC811 and homogenous LiNi_{0.95}Mn_{0.05}O₂ (not to be confused with Ni94) as a function of voltage. The “H2-H3 phase transition remnant” peaks at around 4.2 V become sharper in intensity as the Ni content increases. The capacity in this region also contributes more to the overall specific capacity as the Ni content increases. In addition, the unit cell volume experiences a dramatic decrease as the materials are charged through this peak in dQ/dV, especially for LiNi_{0.95}Mn_{0.05}O₂. This shows that in order to access more capacity for Ni-rich materials, a large lattice volume change will be inevitable.

Figure 2 shows the amount of gas generated during formation (left column) and the charge-transfer resistance R_{ct} measured after formation (right column) as a function of five electrolyte blends for the SC532, SC811 and the core-shell Ni94 pouch cells. The formation was done by charging to two different UCVs. The formation current was the same for the two UCVs of each cell type. This means the formation time was slightly longer when charging to a higher voltage.

For the gas generation, the 2FEC1LFO additive combination generates the least amount of gas regardless of which positive electrode material was used in the pouch cells. For each additive, the amount of gas generated in SC532 cells is similar to that in SC811 cells, which suggests the gas generated was primarily a result of SEI formation on the negative electrode. However, for the Ni94 cells with 2VC1DTD, the amount of gas generated was higher when formed to 4.3 V than when formed to 4.1 V. When LFO was added to the electrolyte in the Ni94 cells, the amount of gas generated when formed to 4.1 or 4.3 V was almost the same.

The charge-transfer resistances, R_{ct} , were generally slightly lower when cells were formed to higher UCV for all the positive electrode

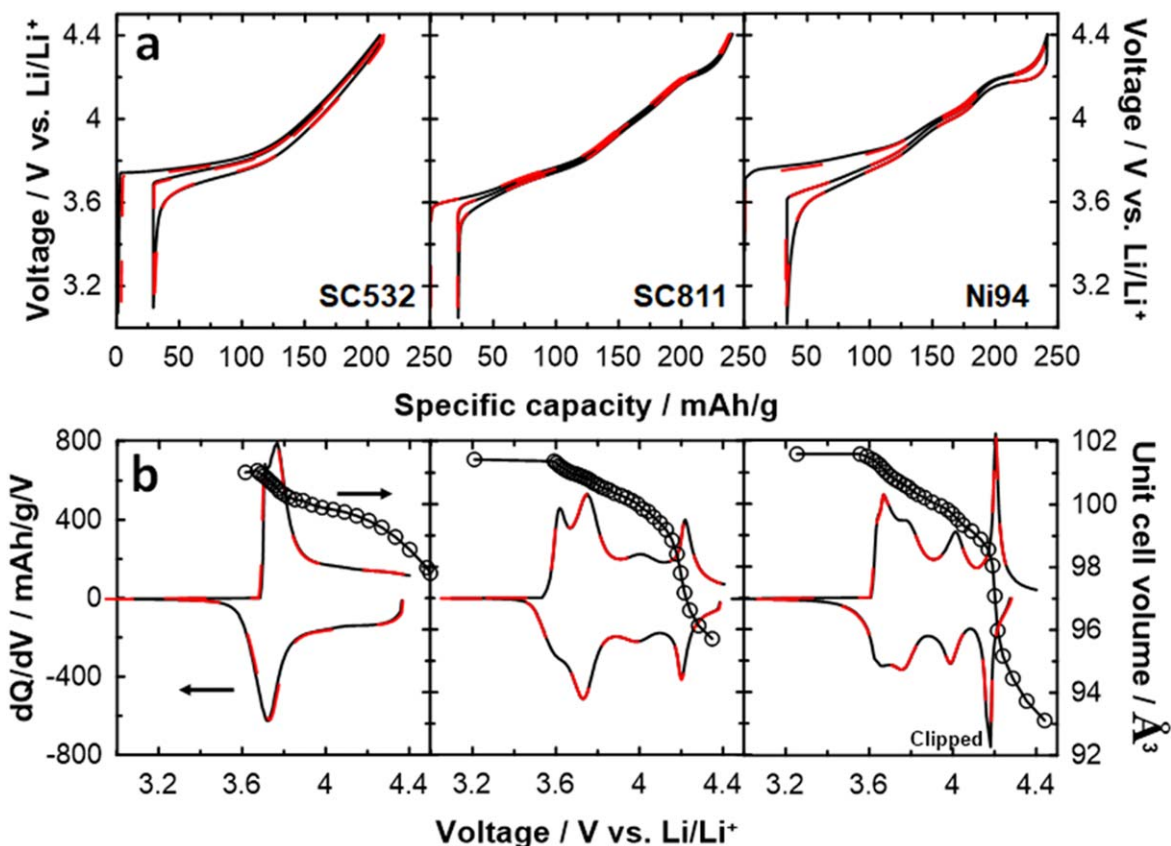


Figure 1. (a) The voltage vs specific capacity of the SC532, SC811 and the prototype Co-free core-shell Ni94 for the 1st 1.5 cycles measured at 30 °C, C/20 from 3.0 to 4.4 V. (b) dQ/dV of the 1st discharge and 2nd charge (left y-axis) and the unit cell volumes (right y-axis) of SC532, SC811 and homogenous LiNi_{0.95}Mn_{0.05}O₂ (not to be confused with Ni94) vs V. The unit cell volume data was taken from Ref. 26.

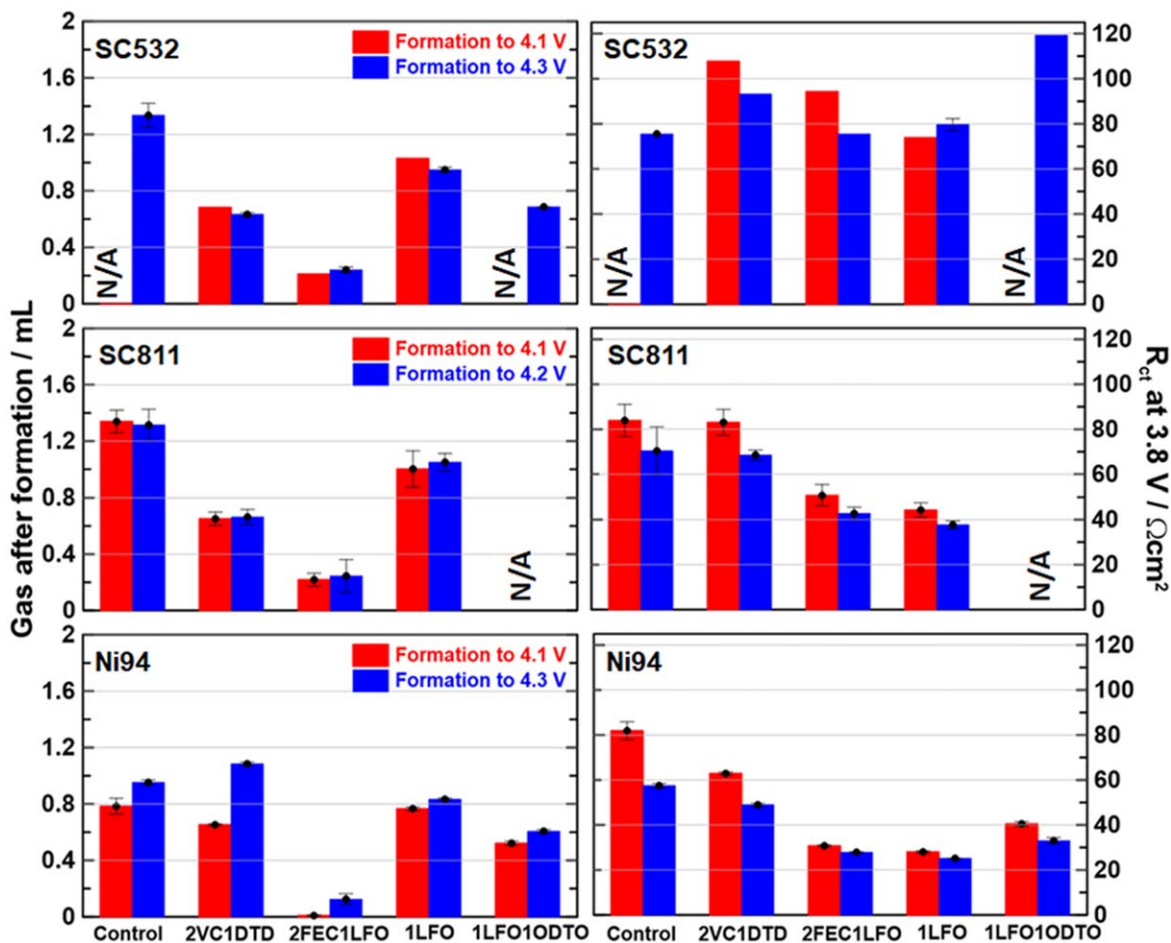


Figure 2. The gas generation (left column) during formation and the charge-transfer resistance R_{ct} (right column) after formation for the SC532, SC811 and core-shell Ni94 pouch cells as a function of electrolyte blends. The data shown without error bars were digitized from J. Li et al.,²¹ L. Ma et al.,²⁷ and X. Ma et al.²² Nyquist plots for SC811 and core-shell Ni94 pouch cells as a function of electrolyte blends are shown in Fig. S1 in the supplementary information.

materials studied here. It is not very clear to the authors at this moment why charging to a high voltage lowers the R_{ct} value. When a full cell charges to a higher voltage during formation, the negative electrodes reach a lower voltage vs Li/Li^+ which may impact the impedance of the SEI formed.³² The incorporation of LFO to the electrolyte generally causes a reduction in R_{ct} compared to cells with 2VC1DTD as has been previously reported.^{2,33,34} Nyquist plots of SC811 and core-shell Ni94 pouch cells measured at 3.8 V and 10 °C after formation are shown in Fig. S1 (available online at stacks.iop.org/JES/169/040531/mmedia).

Figure 3 shows the cell balancing of the SC532, SC811 and the core-shell Ni94 cells as determined by the dV/dQ analysis method.³⁵ The goodness-of-fit of the dV/dQ method is shown qualitatively in Fig. S2. The corresponding positive electrode voltages are labeled when the full cell is charged to 4.1 V and 4.3 V for SC532 (a), 4.06 V and 4.2 V for SC811 (b), as well as 4.04 V and 4.18 V for the core-shell Ni94 (c). The inset images show dQ/dV vs V of the corresponding positive electrode and the dashed lines indicate the corresponding low and high UCVs for the positive electrode vs Li/Li^+ . The “H2-H3 remnant” peaks of the positives, bounded by the dashed lines in the inset dQ/dV curves, are excluded when the full cells are charged to 4.06 V for SC811 and 4.04 V for Ni94. Please note that for SC532 cells, the “H2-H3 remnant” is not observed even at 4.3 V.

Figure 4 shows the charge-discharge cycling performance of the SC532, SC811 and the core-shell Ni94 pouch cells when the UCVs include the “H2-H3 remnant” for SC811 and Ni94. Both the SC811 and Ni94 pouch cells show poor capacity retention without electrolyte additives (control electrolyte). With the help of additives, the

performance of SC811 cells was significantly improved and the core-shell Ni94 cells reach 200 cycles at 80% retention. The capacity retention decreased as Ni content increased with 2VC1DTD, 2FEC1LFO and 1LFO additives. The SC532 cells charged only to 4.3 V do not pass through a large volume change region (Fig. 1b) and they show excellent capacity retention and very little impedance growth. Both the SC532 and SC811 are very robust to particle microcracking, so the differences in capacity retention and voltage hysteresis could be a result of parasitic reactions between the positive electrode material and the electrolyte.¹⁵ Jung et al.³⁶ showed that oxygen release from $\text{LiNi}_x\text{Mn}_y\text{Co}_z\text{O}_2$ positive electrode materials causing chemical oxidation of the electrolyte occurs coincidentally with the large volume contraction which occurs near the “H2-H3 remnant.” This reaction will occur in the case of SC811 but should not occur in the case of SC532. For the polycrystalline core-shell Ni94, Y. Liu et al.⁴ have shown in coin cells that though the high Ni-content core (nominally 100% Ni) showed cracking when cycled up to 4.3 V vs Li/Li^+ for 100 cycles, the shell (nominally 80% Ni) did not. This material was shown to have comparable capacity retention with SC811 in half coin cells. However, the Ni94 pouch cells have a very poor capacity retention and the rate of voltage polarization growth is enormous compared to the SC811 and SC532 cells. The authors speculate this may be related to impact of standard industrial processing on the characteristics of this material. The details of how the Ni94 was processed were not revealed to the authors.

Figure 5 shows the cycling performance of SC532, SC811 and the core-shell Ni94 pouch cells when the UCVs were chosen to exclude the “H2-H3 remnant.” The most obvious difference between

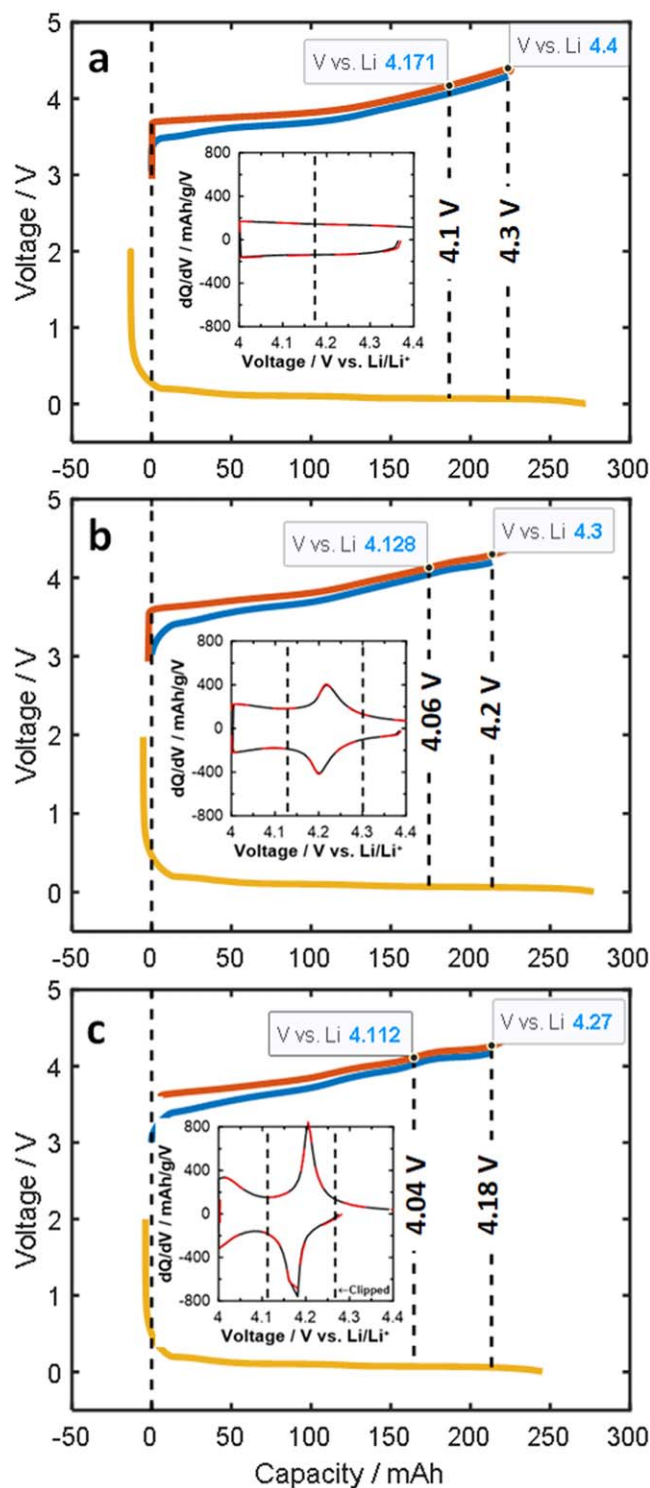


Figure 3. Pouch cell electrode balances for the SC532 (a), SC811 (b) and the core-shell Ni94 (c) measured on the 2nd charge. The dashed lines indicate the two UCVs of the full cells with the positive electrode voltage labeled. The dashed lines in the inset indicate where the labeled positive electrode voltages are in the dQ/dV vs V plots. The goodness-of-fit is shown in Fig. S2.

Figs. 4 and 5 is the vastly improved capacity retention for the Ni94 cells charged to only 4.04 V as well as a dramatic reduction in the rate of growth of voltage polarization. Similarly, the capacity retention and impedance growth for the SC811 cells charged to 4.06 V in Fig. 5 are vastly improved compared to Fig. 4 when the

cells were charged to 4.2 V. These comparisons highlight the fact that the “H2-H3 remnant” is detrimental to cells that require extremely long lifetime. Even a material like the Ni94 used here which capacity degrades very rapidly when used in cell operated to 4.18 V (charging through the “H2-H3 remnant” on every cycle) is relatively acceptable when operated to only 4.04 V by avoiding the “H2-H3 remnant.”

One interesting thing to note is that penalty in absolute cell capacity which occurs by reducing the UCVs from 4.3 to 4.1 V for SC532, from 4.2 to 4.06 V for SC811 and from 4.18 to 4.04 V for Ni94. The cell capacity decrease is the largest for the core-shell Ni94 cells. This is because the “H2-H3 remnant” contributes a larger portion of its capacity, so avoiding this remnant causes a dramatic capacity reduction. This suggests that the longest-lived cells, operated to avoid the “H2-H3 remnant,” with the highest capacity will not be made with positive electrode materials with extremely high Ni contents.

Figure 6 shows the cycling performance for pouch cells with the 2FEC1LFO and 1LFO electrolyte blends at 20 °C when the “H2-H3 remnant” was either included (a) (note that SC532 cell has no “H2-H3” remnant even at 4.3 V) or excluded (b). The cycling performance of the core-shell Ni94 was significantly improved at 20 °C compared to cycling at 40 °C for both UCVs. The capacity retention with 4.18 V UCV is almost tripled at 20 °C compared to at 40 °C. For both UCVs, the significantly lower voltage polarization increase with cycle numbers at 20 °C compared to 40 °C highlights the strong dependency of the side reactions between the electrodes and the electrolyte on temperature. The Ni94 cells charged to only 4.04 V at 20 °C operate without capacity loss and the impedance growth is equal to that of the SC811 cells. Again, by comparing the difference of cycling performance in Figs. 6a and 6b, it is striking how much the capacity retention can be improved by avoiding the “H2-H3 remnant” especially for Ni94 cells.

Figure 7a shows two negative electrodes extracted from core-shell Ni94 pouch cells with control electrolyte or 2FEC1LFO electrolyte blend that were cycled at 40 °C, C/3 CCCV with 4.18 V UCV. The charge-discharge cycling performance of these cells is shown in Fig. 4. The control and 2FEC1LFO cells were cycled to the 290th and 270th cycle, respectively, before disassembly at 4.18 V. Both cells were slowly charged to 4.18 V and voltage hold at 4.18 V for 30 h before opening in glove-box. Both negative electrodes showed a uniform gold color indicating uniform lithiation,³⁷ even though the pouch cell with control electrolyte experienced worse capacity loss compared to the cell with the 2FEC1LFO additive. By the look of the graphite negative electrodes, there appears to have been no significant mass and lithium inventory loss, which also implies there is no significant SEI thickening. Therefore the impedance growth from the negative side is expected to be low and the fast capacity fade of the core-shell Ni94 cell is most likely due to the impedance growth from the positive electrode. The different performance of the cells with control and 2FEC1LFO electrolyte blends can be a result of the different ability to inhibit the impedance rise of the positive electrode by the electrolyte.

Figure 7b shows the cross-section SEM images of a fresh electrode from a dry core-shell Ni94 pouch cell. The SEM images show many of the particles have been fractured due to the calendaring process. Figures 7c and 7d show cross-section SEM images of positive electrodes from two cycled pouch cells with control electrolyte. The pouch cell disassembled in Fig. 7c was cycled to 4.18 V UCV for 290 cycles at 40 °C. This is the cell with control electrolyte shown in Fig. 4 and its negative electrode was examined in Fig. 7a. The pouch cell disassembled in Fig. 7d was cycled to 4.04 V UCV for 200 cycles at 40 °C. This is the cell with control electrolyte shown in Fig. 5. Particles in both electrodes show microcracking. Even though the in situ XRD in Fig. 1 and many other reports in the literature indicate the generation of microcracking when a polycrystalline material was repeatedly cycled to high voltage,^{13,16,38} it is difficult from these SEM images to tell whether the microcracks were generated due to initial calendaring or induced by anisotropic

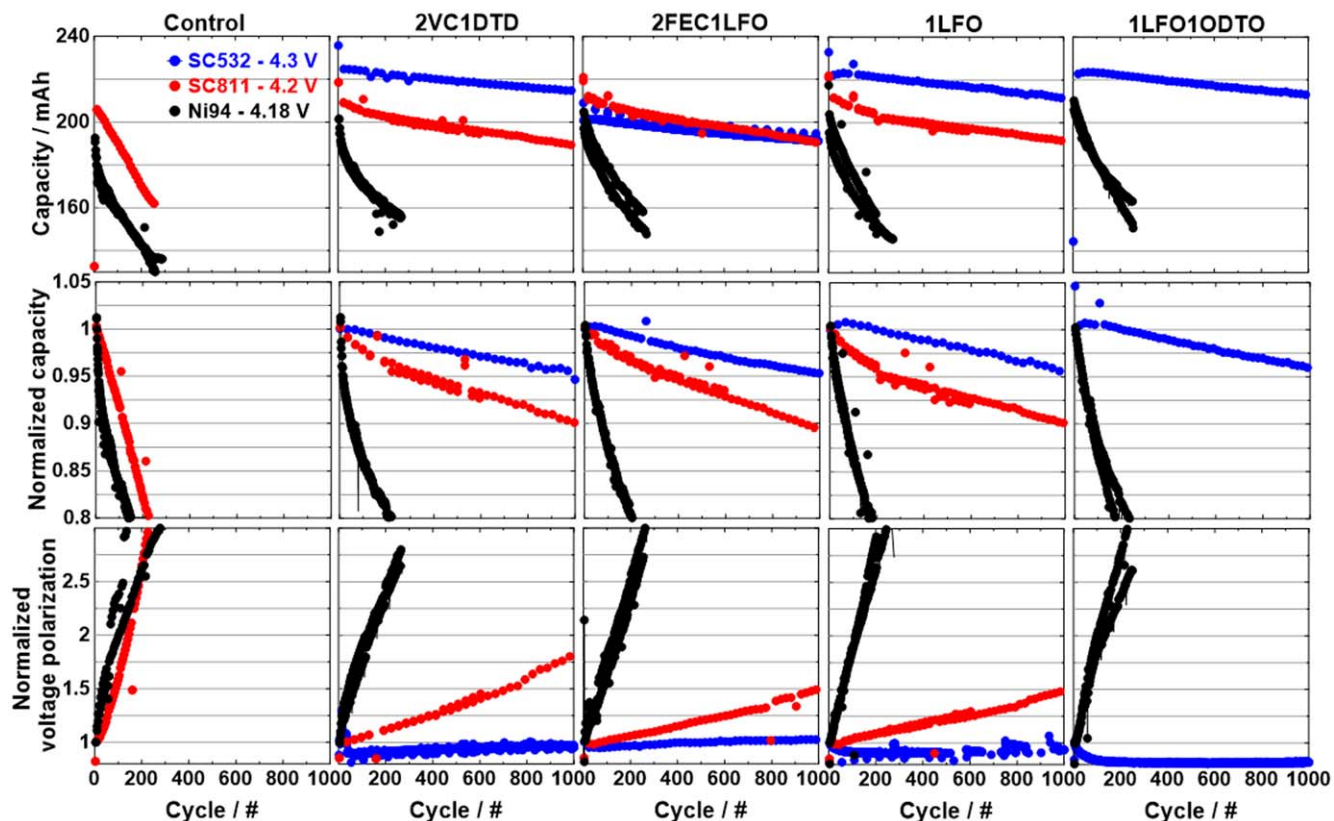


Figure 4. Capacity, normalized capacity and normalized voltage polarization vs cycle for five electrolyte blends measured at 40 °C, C/3 CCCV when the UCVs include the “H2/H3 remnant” for the case of SC811 and Ni94. The 5th cycle is chosen as ‘1’ for the normalization.

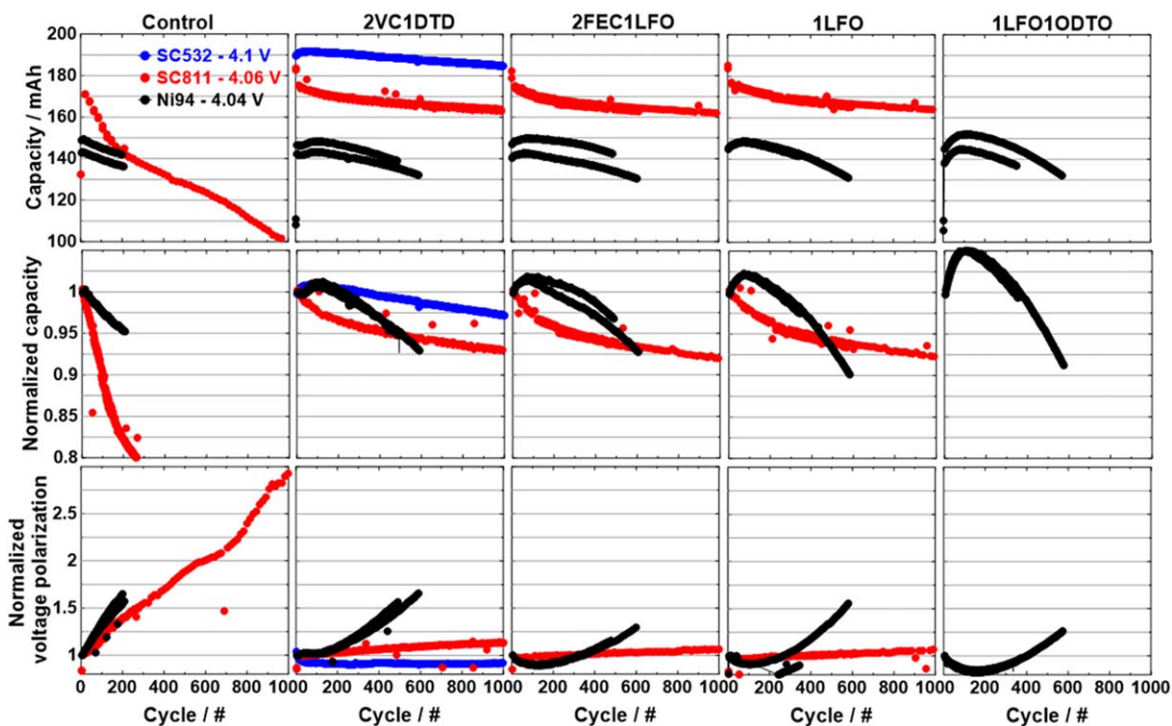


Figure 5. Capacity, normalized capacity and normalized voltage polarization vs cycle for five electrolytes measured at 40 °C, C/3 CCCV when the UCVs exclude the “H2/H3 remnant.” The 5th cycle is chosen as “1” for the normalizations.

volume change from repeatedly cycling to a high voltage. The goal of using a core-shell structure is to protect the Ni-rich core from the electrolyte exposure using a low Ni content shell so that both high

capacity and long cycle life can be achieved. Therefore, the microcracks due to calendaring are particularly detrimental for the core-shell Ni94 material as the Ni-rich core will have a direct

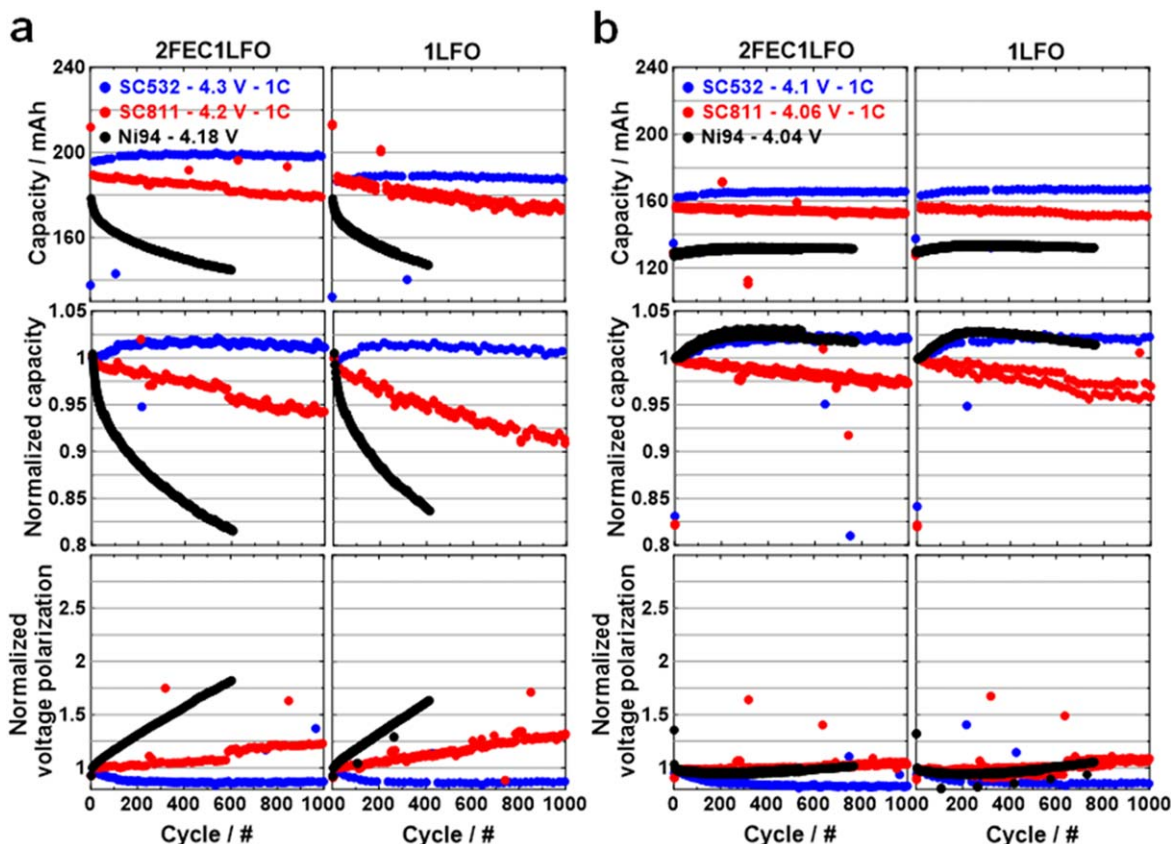


Figure 6. Capacity, normalized capacity and normalized voltage polarization vs cycle for 2FEC1LFO and 1LFO electrolyte blends measured at 20 °C when the UCVs include (a) and exclude (b) the “H2/H3 remnant.” The 5th cycle is chosen as “1” for the normalizations. The SC532 and SC811 cells were measured with 1C CCCV. The core-shell Ni94 was measured with C/3 CCCV.

exposure to the electrolyte which is expected to cause severe side reactions during cycling. It was shown previously that the SC532 and SC811 fresh electrodes have minimal microcracking from calendaring, which highlights the advantageous processing ability of single crystal materials compared to the polycrystalline ones.¹⁵

Figure 8a shows the C/20 and C/5 dQ/dV curves of the core-shell Ni94 and the SC811 before and after cycling from 3 to 4.3 V vs Li/Li⁺ in coin cell half-cells at 30 °C. The C/20 dQ/dV curve of the core-shell Ni94 remains stable only up to ca. 4.1 V, whereas that of the SC811 is virtually unchanged over the entire voltage range. The significant change of the core-shell Ni94 dQ/dV curve occurs around the so-called “H2-H3 remnant” at the 4.2 V dQ/dV region where a large volume contraction occurs according to in situ XRD. Half-cell cycling with C/20 ensures that impedance contributions to capacity fade are negligible. If there is any portion of the active material permanently disconnected from the conducting network, one would expect the entire area enclosed by the dQ/dV curve to decrease by that portion, instead of only influencing the “H2-H3 remnant” region. It is therefore reasonable to believe that it is only a temporary active material disconnection from the conducting network due to the lattice volume contraction over the “H2-H3 remnant” region.³⁹ Once the material expands again, it is reconnected to the conducting network so that the dQ/dV curve below 4.1 V remains stable. On the other hand, the SC811 half cell shows virtually no changes around its 4.2 V peak region, which may imply all the active particles are well-connected to the conducting network even though it also experiences a large lattice volume change over this region. This difference from the core-shell Ni94 is most likely due to the single crystal structure without microcracking that helps minimize electric-disconnection over the high voltage range, whereas, for the polycrystalline core-shell Ni94, the microcracking by electrode calendaring might even exacerbate the electric-

disconnection. In the C/5 cycling case of the core-shell Ni94 half cell, the 4.2 V peak from charging almost entirely vanishes after cycling and even the middle range of the dQ/dV curve has shifted in voltage which indicates a large impedance effect. The SC811 shows very stable C/5 cycling over the entire voltage range. The coin cell half-cell cycling data is shown in Fig. S3.

Figure 8b shows the C/3 CCCV cycling performance of a core-shell Ni94 pouch cell tested with alternating UCVs at 40 °C and 20 °C. When cycling with a 4.18 V UCV, the cell capacity decreases significantly faster compared to when the UCV is 4.04 V. In fact, at 40 °C, the capacity vs cycle number segments for the 4.04 V UCV are virtually horizontal and all capacity loss occurs during the 4.18 V cycling. The situation is similar at 20 °C. This behavior strongly suggests a recommended use mode for cells with any high Ni content positive electrode. If a large discharge capacity is not required, charging to a voltage below the “H2-H3 remnant” will help preserve cell life time. The user should only charge to a voltage beyond the “H2-H3 remnant” in cases when a large discharge capacity is needed.

Conclusions

The cycling performances of SC532, SC811 and a prototype polycrystalline Co-free core-shell Ni94 material were compared in pouch cells with artificial graphite negative electrodes. Two upper cut-off voltages (UCVs) were carefully chosen for the SC811 and Ni94 cells based on the dQ/dV curves of the positive electrode material. The lower UCV was chosen to exclude the so-called “H2-H3 phase transition remnant” which is around 4.2 V in the dQ/dV vs Li/Li⁺ curve, whereas the higher UCV was chosen to include this “remnant.”

For all five electrolyte blends tested, the SC532 cells showed the best capacity retention with both UCVs and at both temperatures.

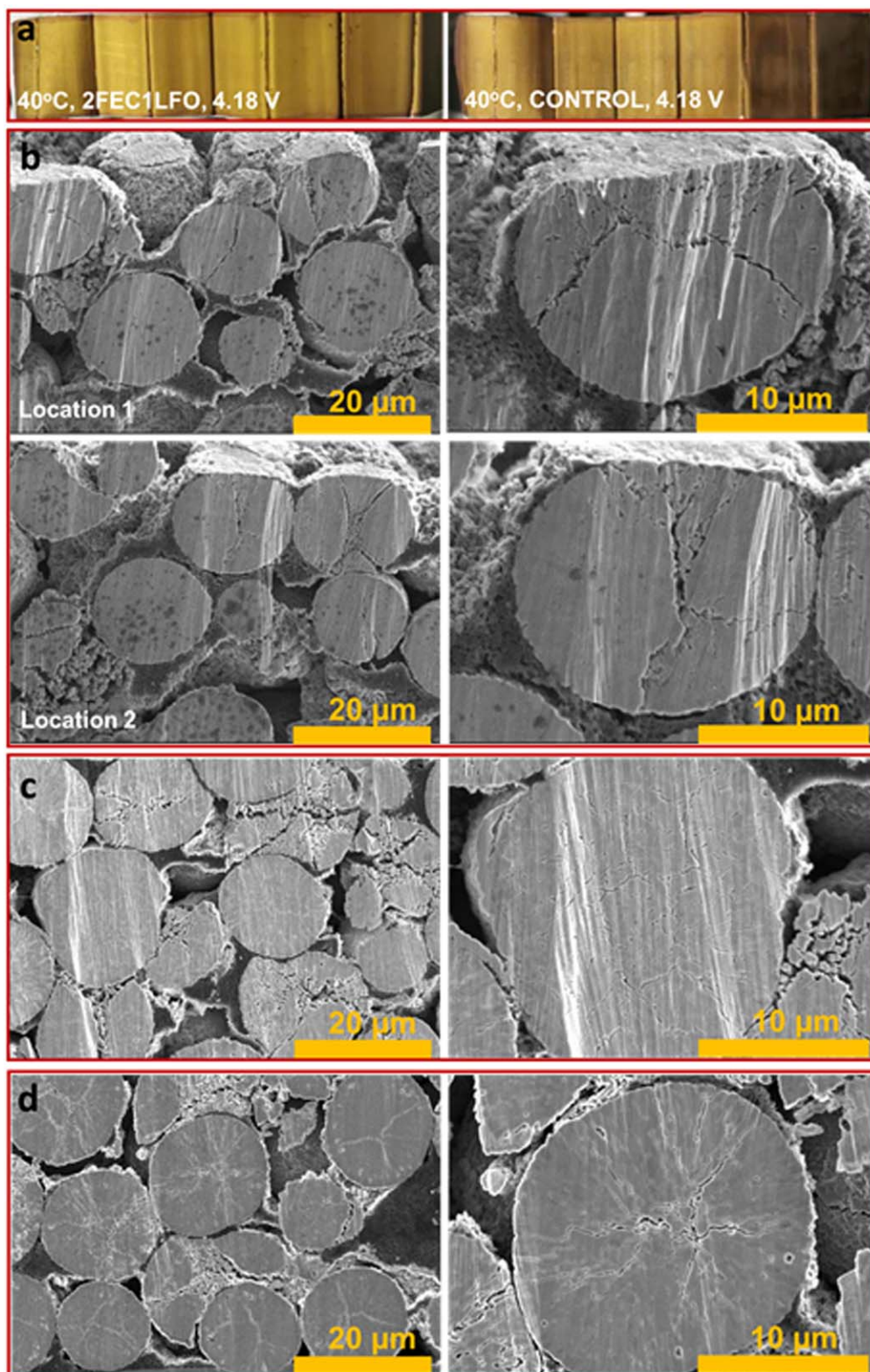


Figure 7. (a) The harvested graphite negative electrodes at 4.18 V from pouch cells cycled with control (290 cycles) and 2FEC1LFO (270 cycles) electrolyte at 40 °C, C/3 CCCV. (b) Cross-section SEM images of two locations of a fresh core-shell Ni94 electrode punched from a dry pouch cell. (c) Cross-section SEM images on a cycled Ni94 electrode with a 4.18 V UCV and control electrolyte (290 cycles) at 40 °C (d) Cross-section SEM images a on cycled Ni94 electrode with a 4.04 V UCV and control electrolyte (200 cycles) at 40 °C.

SC811 cells have lower lifetime than that of SC532 cells but better lifetime than the prototype polycrystalline Co-free core-shell Ni94 at both 20 °C and 40 °C. The core-shell Ni94 only has excellent life time at 20 °C, 4.04 V UCV which is comparable to SC532 and

SC811, at least for the first 800 cycles shown in this work with C/3 CCCV charge-discharge cycling.

The Ni94 cells showed poor capacity retention and large impedance growth when operated to 4.18 V (cycling through the

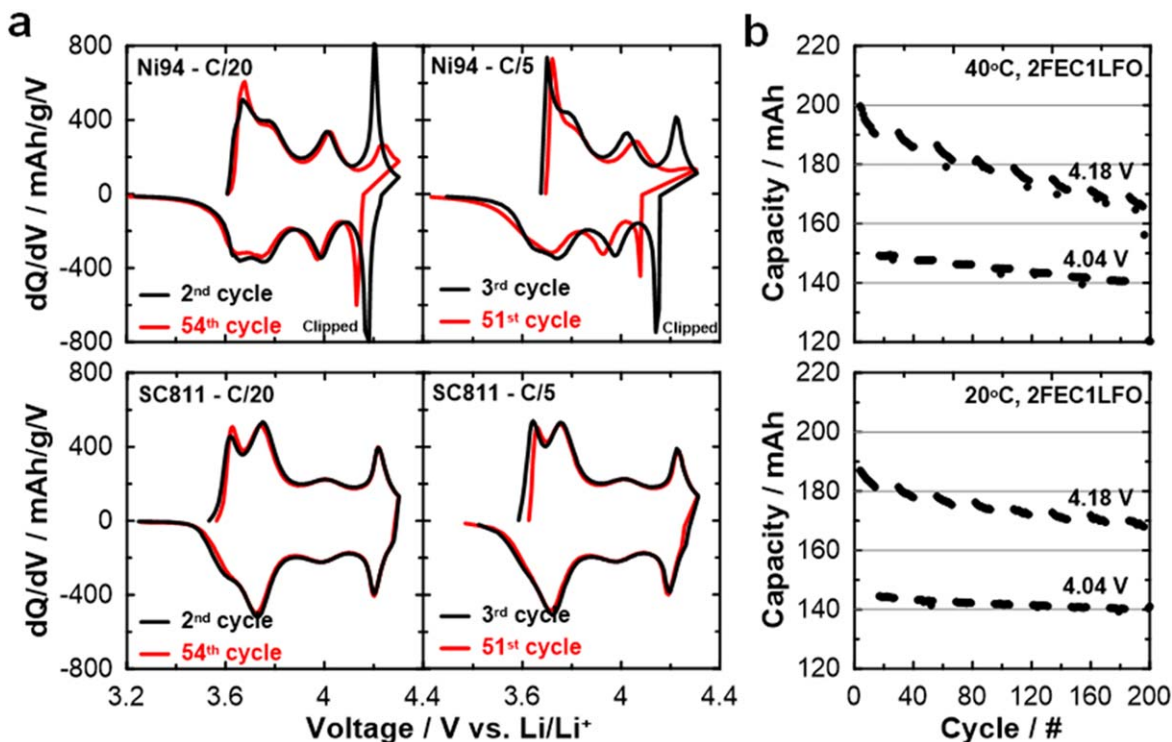


Figure 8. (a) The C/20 and C/5 dQ/dV curves of the core-shell Ni94 and the SC811 before and after cycling from 3 to 4.3 V vs Li/Li⁺ in coin cell half-cell at 30 °C. (b) The capacity vs cycle number of core-shell Ni94 pouch cells tested with alternating 4.18 V and 4.04 V UCVs at 40 °C and 20 °C, C/3 CCCV.

“H2-H3 remnant”). However, the performance was dramatically improved, even for this apparently poor material, by limiting the upper cut-off to 4.04 V to avoid the “H2-H3 remnant.” This suggests a universal strategy for obtaining the longest cell life time with high Ni-content positive electrode materials. Unless a large discharge capacity is required, limiting the charge cut-off voltage below the “H2-H3 remnant” lowers capacity degradation significantly. Figure 8b explored this scenario and showed that capacity loss only occurred during the 4.18 V UCV cycles.

For the prototype core-shell Ni94, the cross-section SEM images on fresh electrodes have shown microcracks due to electrode calendaring. These microcracks are particularly detrimental to cell cycle life as the exposed Ni-rich core is expected to be very reactive with the electrolyte to induce large impedance growth from electrolyte oxidation. Half cell cycling and pouch cell cycling with alternating UCV showed the temporary loss of electric-connection of the active material happens when cycling over the so-called “H2-H3 phase transition remnant” region of the polycrystalline core-shell Ni94 positive electrode material. A large impedance increase is induced at the same time.

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